

Question 01: The Turn-Based Battle Simulator

1. Scenario

You are building the combat engine for a retro RPG game. A player encounters a Goblin! The player needs to decide whether to attack, heal, or run away. The Goblin will always attack back unless it is defeated.

2. Technical Requirements (The "Rules")

1. **Variables (State Management):**
 - `player_hp`: Start at 50.
 - `monster_hp`: Start at 50.
 - `potions`: Start at 3 (Used to heal).
2. **The Main Loop:**
 - The game runs inside a loop that continues as long as **both** `player_hp > 0` AND `monster_hp > 0`.
3. **The "Visual":**
 - **The Twist:** Every time the loop restarts to show the menu, print a "Health Bar" for the player.
 - *Example:* If player has 30 HP, print 3 symbols (like `###`) to represent their health graphically (e.g., 1 `#` per 10 HP).
4. **Conditionals:**
 - **Option 1 (Sword Attack):** Subtract 10 from `monster_hp`.
 - **Option 2 (Heal):** if `potions > 0` add 15 to `player_hp` and subtract 1 potion. *otherwise*, print "No potions left!"
 - **Monster Logic:** After the player moves, if the monster is still alive (`monster_hp > 0`), subtract random damage (e.g., 5-15) from the player.
5. **Exception Handling:**
 - If the user enters a non-integer (like "attack"), catch the error and force them to choose again without the monster attacking them that turn.

3. Sample Expected Output

None

--- BATTLE START ---

Player Health: ##### (50)

Monster Health: 50

Potions left: 3

1. Sword Attack

2. Drink Potion

3. Run Away

Choose action: 1

> You hit the monster for 10 damage!

> The monster attacks you back for 12 damage!

Player Health: ### (38)

Monster Health: 40

Potions left: 3

Choose action: 2

> You drank a potion. Recovered 15 HP.

> Potions remaining: 2

> The monster attacks you back for 8 damage!

Player Health: ##### (45)

Monster Health: 40

...

Choose action: attack

> Invalid Input! Please enter 1, 2, or 3.

...

(Game ends when someone hits 0 HP)

Question 02: The Debt Crusher (Amortization Calculator)

1. Scenario

You are building a financial tool for a bank. A customer wants to know: 'If I owe \$5,000 and pay \$200 a month, when will I be debt-free?' Your program needs to simulate the payment process month by month until the balance reaches zero.

2. Mathematical Concept

You must understand that Interest is calculated every month on the current balance

$$\text{MonthlyInterest} = \text{CurrentBalance} \times \left(\frac{\text{AnnualInterestRate}}{12} \right)$$

$$\text{NewBalance} = \text{CurrentBalance} + \text{MonthlyInterest} - \text{MonthlyPayment}$$

3. Technical Requirements

1. Variables & Inputs:

- Ask for **principal** (Total debt)
- Ask for **annual_rate** (e.g., 5 for 5%). Convert this to a decimal (0.05) and then to a monthly rate (0.05 / 12)
- Ask for **monthly_payment**

2. Exception Handling:

- Users might type "5%" instead of "5" or use currency symbols. Catch these errors and ask for clean numbers.

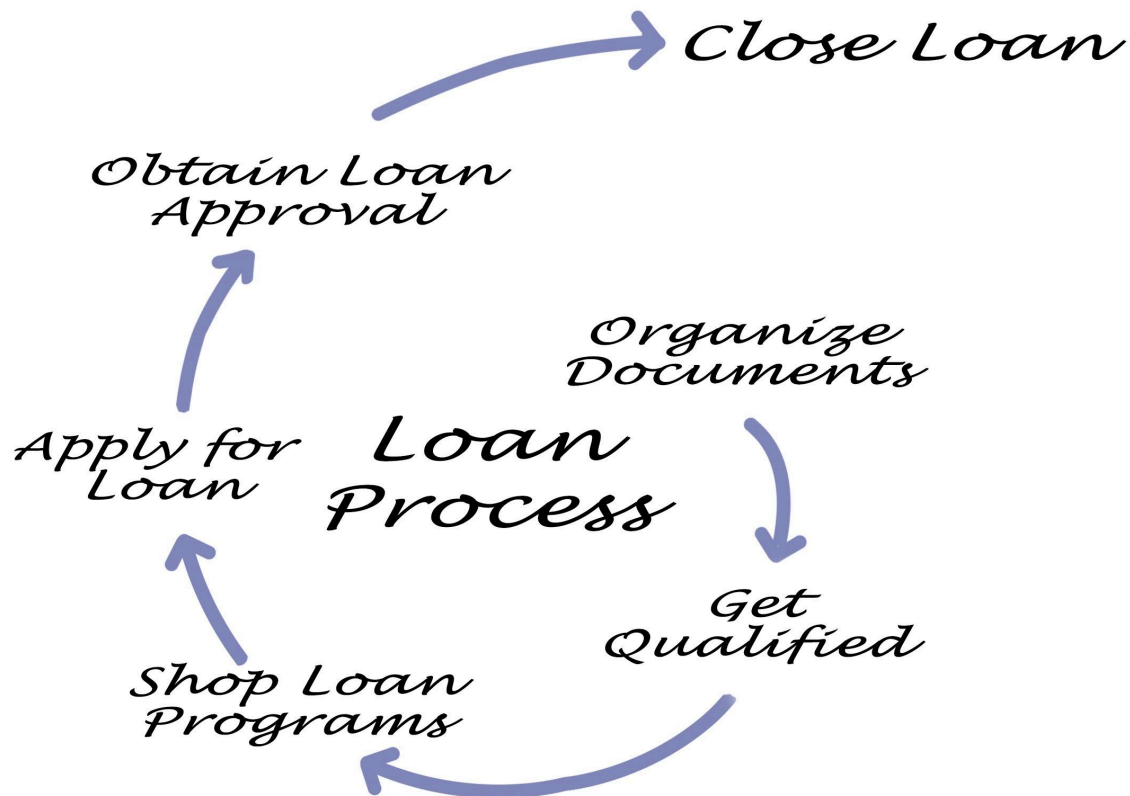
3. The Logic Trap:

- **Critical Step:** Before starting the loop, check if the **monthly_payment** is *less* than the first month's interest.
- *Why?* If Interest is \$50 and Payment is \$40, the balance grows forever. The program must detect this and print: "You will never pay this off!" and stop.

4. The Loop:

- Run **until balance is 0**
- Inside the loop:
 - Calculate interest for the current month.

- Calculate how much of the payment goes to principal ($\text{payment} - \text{interest}$).
- Subtract that principal portion from the balance.
- Increment a **month** counter.
- **The Final Month Twist:** If the remaining balance is *less* than the monthly payment, do not subtract the full payment! Only pay what is left.



4. Sample Expected Output

None

```
--- LOAN CALCULATOR ---
```

```
Enter Loan Amount: 1000
```

```
Enter Annual Interest Rate (%): 12
```

```
Enter Monthly Payment: 300
```

```
Calculating...
```

Month 1:

Interest: \$10.00

Principal Paid: \$290.00

Remaining Balance: \$710.00

Month 2:

Interest: \$7.10

Principal Paid: \$292.90

Remaining Balance: \$417.10

Month 3:

Interest: \$4.17

Principal Paid: \$295.83

Remaining Balance: \$121.27

Month 4:

Interest: \$1.21

Principal Paid: \$121.27 (Final Payment)

Remaining Balance: \$0.00

TOTAL: Paid off in 4 months.

Total Interest Paid: \$22.48